

An edge-first federated architecture: Model 1 registry as control plane, Model 2 feed access, Model 3 adapter spine.

80

CAMERAS ONBOARDED

26 ms

VEHICLE TRACE

1.04 s

DETECTION → ALERT

869

AUTOMATED TESTS

Working platform. Every figure in this deck is measured on the running system, or arithmetic from a measured figure with the calculation shown.

THE CONSTRAINT THAT SHAPES EVERYTHING

“Participants will be provided with a designated vehicle registration number.”

By the time that number is handed over, **the vehicle has already driven past the cameras.** There is no opportunity to go and look for it.

The only thing that can answer is an index of **every plate already read.**

CONSEQUENCE

We persist every read, not just watchlist hits

This moves the system's cost from the query path to the write path.
Every sizing decision in this deck follows from it.

GJ 01 AB 1234

The distinction that matters: a watchlist alerter answers questions asked in advance. An index answers a question nobody had asked yet, which is exactly what the evaluation is.

PROPOSED MODEL, WITH JUSTIFICATION

Hybrid: a registry control plane, direct feed access, an adapter spine

We adopt **Model 4's analytics outcomes** through distributed edge inference rather than centralised video ingest.

MODEL	ROLE IN THE HYBRID	DEMONSTRATED BY
M1: Registry & GIS mandatory	The control plane, not a side deliverable. Adapter choice, RBAC scope, health, coverage and route plausibility all resolve through it	80 cameras onboarded (50 simulated, 30 government); GIS map; corridor gap analysis
M2: Unified viewing	The feed-access and operator layer. Direct RTSP/ONVIF is genuinely right for cameras with no VMS in front of them, which is most of them	Live view, searchable sightings, alert console
M3: Federation	The spine that makes M2 survive scale. A new vendor is a plugin, not a redesign	One interface; adding a source type changes no other file
M4: Analytics outcomes	Its goals adopted in full. Its centralised ingest rejected on cost and on the Core Goal	Journey reconstruction, watchlist correlation, real-time alerts

The **organising idea**: move the analytics to the edge, move only metadata to the centre, move video only on demand.

PROPOSED MODEL, WITH JUSTIFICATION

Why we rejected Model 4, and the arithmetic at 80,000 cameras

	CENTRAL VMS	THIS PLATFORM
Sustained backhaul	240 Gbit/s	~40 Mbit/s
Central storage, 15 days	~26 PB	~5 GB/day
Existing VMS, storage, AMC	Replaced	Preserved
On WAN outage	Blind	Edge keeps detecting

0.5 kbit/s per camera is measured, not assumed, 104 sightings/minute across 71 contributing cameras, at ~300 bytes a row.

REASON ONE

It contradicts the Core Goal

The brief asks for an approach that is *cost-effective* and *uses existing infrastructure to the maximum practical extent*. Model 4 discards 26 departments' investments and rebuilds them centrally.

REASON TWO

It contradicts the bonus criteria

Which explicitly reward *strong edge-processing, bandwidth-optimisation, or low-connectivity operation*. Central ingest is the opposite of all three.

We decline one implementation choice, not a capability. Every functional outcome Model 4 lists is present. The transport is what differs.

SOLUTION OVERVIEW AND OBJECTIVES

Three capabilities, graded separately, all built

OBJECTIVE	WHAT IT MEANS	STATE
Onboard	Any camera, any brand, any protocol, without redesign	80 cameras. 5.1 s each, against a 30 s budget
Read	Continuously read plates from every camera and store all of them	82.4% exact on ANPR-grade cameras · 104 sightings/min
Trace & alert	A plate returns a route across the state; a wanted vehicle raises an alert in real time	26 ms trace · 1.04 s detection→alert

ALSO BUILT, AND ALSO GRADED

Evidence export

CSV and PDF carrying plate, confidence, camera, geolocation, timestamp **and the crop that proves the read.**

Coverage & gap analysis

Which road segments no camera watches, the input to where the next cameras go.

Audit & access control

Role- and department-scoped access, with an immutable record of every search and export.

KEY INNOVATIONS

Five, each backed by a measurement rather than a claim

	INNOVATION	THE MEASUREMENT BEHIND IT
1	Edge-first metadata architecture , analytics beside the camera, ~250-byte events northbound	0.5 kbit/s per camera against ~3 Mbit/s of video, four orders of magnitude
2	Registry as control plane , not inventory, adapters, RBAC, health, geography and route plausibility all resolve through it	The organisers' grid changed hostname mid-build. It cost one environment variable
3	Per-track confidence voting + derived confusable-character normalisation , the confusion table is generated, never typed in	Coercion is positional ; a hand-written table cannot distinguish a considered omission from an oversight
4	Travel-time plausibility scoring , an impossible hop splits the journey instead of being dropped	Cloned-plate detection as a by-product of route reconstruction
5	Self-calibrating tamper detection , each camera judged against its own normal, not a global threshold	False positives 39 → 0

Six layers, and one of them is mandatory by design

L5	PRESENTATION	GIS console · Alert desk · Journey explorer · Registry
L4	INTELLIGENCE	Watchlist matcher · Journey · Re-ID · Reports
L3	PLATFORM CORE	Event store · ★ CAMERA REGISTRY & GIS · RBAC · Audit · Stream broker
L2	FEDERATION	Adapter framework · Credentials · Health prober
L1	EDGE ANALYTICS	Decode · Sample · Detect · Track · OCR · Vote · Tamper
L0	SOURCES	IP cams · Encoders · NVRs · Departmental VMS

★ mandatory Model 1, as the control plane

WHAT CROSSES THE DISTRICT → STATE LINK

Metadata always. Video only on demand.

A relay starts when an operator opens a camera and is reaped 45 seconds after the last viewer. That cost is **per viewer, not per camera**, an unwatched camera costs nothing beyond its metadata.

DEPLOYMENT TIERS

Edge → district → state

One container image, identical in all three placements. Scaling is a replica count, not a redeployment.

A vehicle passes a camera in Rajkot

- ① registry already knows the camera, its position, and its adapter
- ② stream opened - timing from the video's own **PTS**, never from arrival
- ③ vehicle detected and tracked; plate located **inside the vehicle box**
- ④ OCR on each sampled crop
→ per-track vote → **ONE plate string**
- ⑤ sighting written - **always**, every vehicle, ~300 bytes
└ in the **same transaction**: matched against the watchlist, alert written if it hits. Both rows commit, or neither
- ⑥ operator sees the alert with its evidence bundle **1.04 s**
- ⑦ later - a plate is typed, and its route across the state returns **26 ms**

Steps ⑤ and ⑦ are the product. ⑤ is what makes ⑦ possible on evaluation day.

NOT OBVIOUS, AND LOAD-BEARING

Plate search is confined to a vehicle

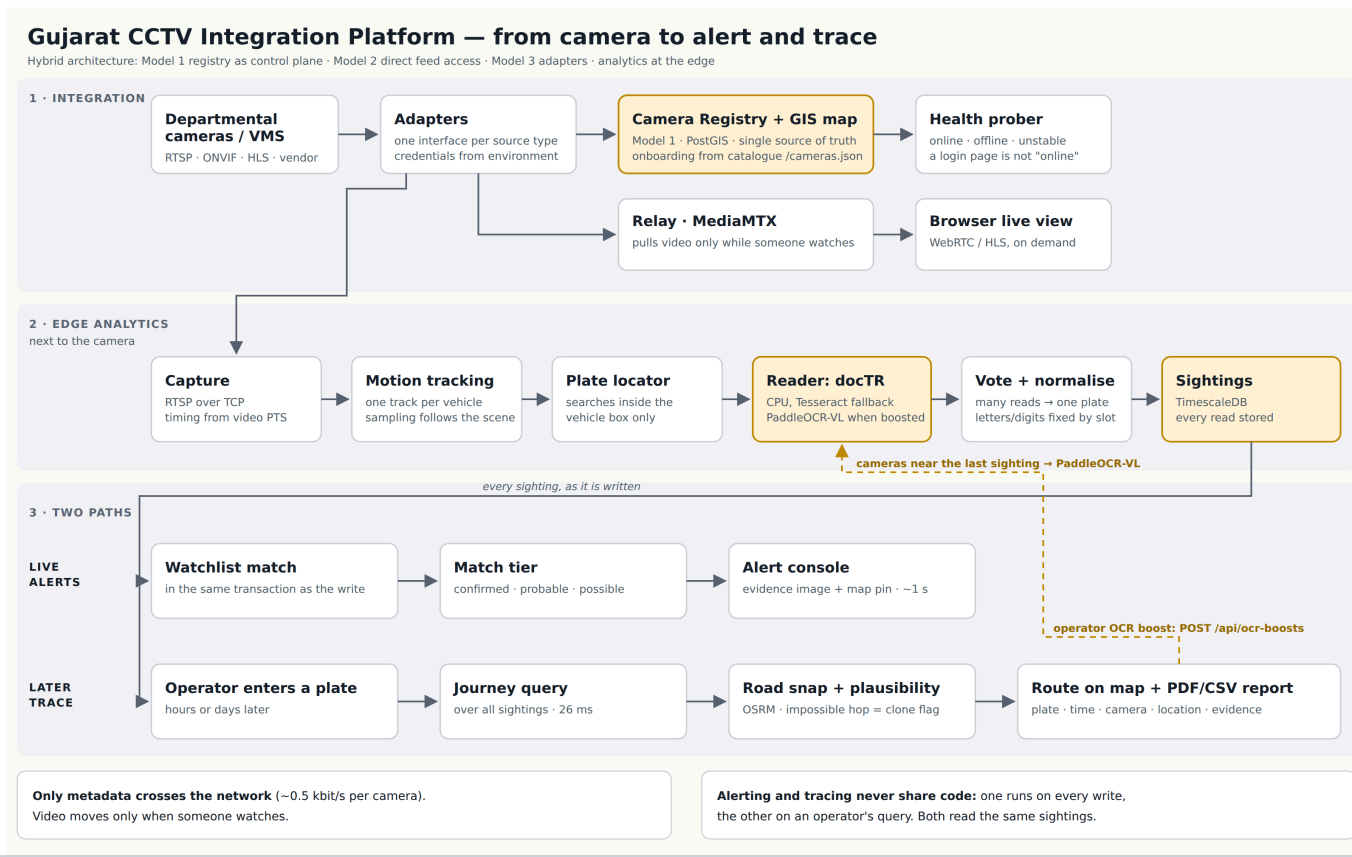
Every government feed burns a timestamp band and site label onto its own picture, at plate size and contrast. Whole-frame recognition reads the camera's own clock as a registration number, on every camera, forever.

NOT OBVIOUS, AND LOAD-BEARING

Timing comes from the video

A gateway replays its buffer on connect, so frames arrive faster than real time. Timed by arrival, **every reconnect computes an impossible speed and reports a cloned plate.**

Three stages: integration, edge analytics, and two paths that never share code



Read the bottom band first. Live alerting runs on every write; the retrospective trace runs on an operator's query. They read the same sightings table and share no implementation, because the evaluation grades both and building one does not deliver the other.

Detection, recognition, and event analytics

decode → **sample** → detect + track → **locate plate** → OCR → reject overlay text → **per-track vote** → normalise → persist

STAGE	APPROACH	WHY IT IS THAT WAY
Sampling	Analysis rate follows scene activity	80,000 × 15 fps = 1.2 M detections/s does not close. 81.3% of analysis avoided
Detect & track	Multi-object tracking, stable id per vehicle	Voting needs a track, not a frame
Plate localisation	Inside the vehicle box only	Removes the burnt-in-overlay error class entirely
Recognition	docTR on the CPU; PaddleOCR-VL on GPU for cameras an operator boosts	Night plates found 35% → 62% → 100% on the test clips
Voting	Whole-string <i>and</i> per-character, confidence-weighted	Per-frame emission duplicates records <i>and</i> discards many independent opinions
Normalisation	Positional , coerced by plate slot, at write <i>and</i> query time	A global 0→0 merges different vehicles onto one key
Persistence	Unconditional, including format failures	A mis-read wanted vehicle is worse than a noisy record

Beyond ANPR: vehicle class and appearance attributes · **a vehicle id on every sighting**, joined by plate or by a learned appearance vector (DINOv2-small), so an unreadable vehicle can still be followed · self-calibrating tamper detection · per-camera health analytics · **on-demand OCR boost** for the cameras around a location.

AI ANALYTICS: WHAT THE MEASUREMENTS SAID

Accuracy is reported per condition, because one headline would average three problems

CONDITION	EXACT	WITHIN 2 CHARS
Daylight	86.6%	98.1%
Glare	57.1%	92.9%
Night	58.6%	75.9%
Overall, ANPR-grade	82.4%	95.1%

Night is the weak spot, and it is on the slide rather than in a footnote. Measured with Tesseract; docTR, the default reader since 14 Sep, finds 62% of night plates against Tesseract's 35% on the same clips.

THE FINDING THAT SURPRISED US

Heavier was not better

Tesseract on Indian plates	83% exact
A general hub OCR model, same crops	1.1% exact
Single-image GPU inference	24.4 ms
The CPU path it was meant to beat	17.0 ms
docTR on the CPU, night plates found	62% vs 35%

Light models by default, heavy models as a measured per-camera fallback. A camera the light path demonstrably cannot read escalates on its own evidence, one rung at a time, budgeted, and rolled back if it does not help.

THE READER LADDER, MEASURED

Night is where readers separate, and it is the condition that matters

READER	NIGHT	DAY
Tesseract	34.7	84.1
docTR PARSeq	62.4	95.7
PaddleOCR-VL	100.0	100.0

Plates found on the 12 generated traffic clips, 101 night and 69 day. Measured 14 Sep 2026. docTR is the default because it fits a CPU edge box at 37 ms per crop; PaddleOCR-VL needs a GPU and runs only on cameras an operator boosts.

Day hides the difference. Every reader is respectable in daylight, so a daylight-only benchmark would have justified the cheapest one and left the estate blind at night, which is when the vehicles we are asked about actually move.

A match is not a boolean

Matching happens **on the write path**, not in a downstream consumer. The watchlist is held in process and refreshed on a 2-second timer, at 104 sightings a minute, a per-sighting query puts a network round trip on the pipeline's hot path.

TIER	CONDITION	WHAT IT MEANS TO AN OPERATOR
confirmed	Exact key and confidence ≥ 0.85	Act on it
probable	Exact key at lower confidence, or edit distance 1	Verify against the evidence crop first
possible	Edit distance 2	A lead, corroborate before acting

WHY NOT EXACT-ONLY

It misses the wanted vehicle whose plate came back one character wrong, **12.7% of reads**. A system that misses one in eight is not working.

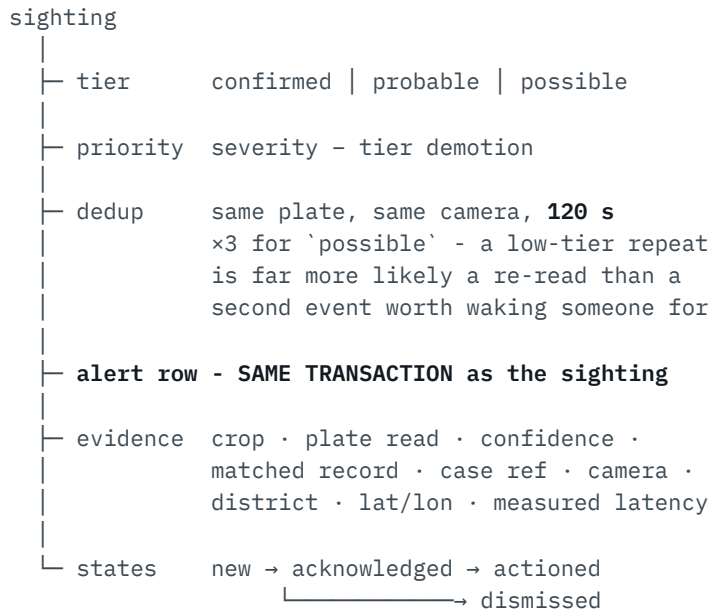
WHY NOT ANYTHING-CLOSE

The operator is flooded and stops reading alerts, which is **worse than not alerting at all**.

WHY CONFIDENCE SPLITS AN EXACT MATCH

The same string at 0.4 and at 0.95 is different evidence. The tier is how the operator is told **which one they are looking at**.

Detection to an operator's screen in 1.04 seconds



1.04 s

MEASURED

5 s

BUDGET

CONSEQUENCE ONE

Both rows commit, or neither does

The time-series store cannot enforce that foreign key, so the transaction enforces it structurally: **an alert can never point at a sighting that does not exist.**

CONSEQUENCE TWO

There is no delete

`dismissed` and `false_positive` are statuses. An alert an operator judged wrong is itself evidence, of the error rate, and of the decision taken.

KEY TECHNOLOGIES, FRAMEWORKS AND TOOLS

Every choice has a reason, and the deviations are named

LAYER	CHOICE	WHY THIS ONE
Store	TimescaleDB (PostgreSQL + PostGIS)	Registry, geospatial and time-series in one database, removes a distributed-consistency problem we do not need
Plate search	pg_trgm	Sufficient at this scale; a search cluster is the scale path, not the starting point
Vectors	pgvector	Vehicle appearance search (DINOv2-small, 384 numbers) inside the database already present
Media	MediaMTX	RTSP in, WebRTC/HLS out, the relay the brief already anticipates
Events	Redpanda	Kafka wire protocol, single binary, no ZooKeeper
Routing	OSRM + Gujarat OSM extract	Road-network snapping and travel-time plausibility
API	FastAPI + OpenAPI	Documented and browsable, integration-ready by construction
Map	MapLibre GL	GPU-rendered. Leaflet's DOM markers do not survive statewide density, and 80,000 points is the claim

Justified deviations from the suggested stack: MapLibre over Leaflet, TimescaleDB + trigram over Elasticsearch, Redpanda over Kafka. Each is stated with its reason, because unexplained deviation is penalised and justified deviation is not.

SCALABILITY AT 80,000 CAMERAS

The sizing number is measured, and so is what breaks it

50 to 57 cameras per 20-core box. At 57 cameras, recognition p50 was **624 ms** with a healthy 6.1% shed. At 76 cameras the same box shed **94.7%** and p50 rose to 3.4 s.

QUANTITY	ARITHMETIC	RESULT
Nodes for 80,000 cameras	$80,000 \div 50$	1,600
Per district	$1,600 \div 33$	~48
Sightings/day	$80,000 \times 1.5/\text{min} \times 1,440$	~173 M
Compressed rows/day	$173 \text{ M} \times 300 \text{ B} \div 10$	~5 GB
Metadata backhaul	$80,000 \times 0.5 \text{ kbit/s}$	~40 Mbit/s

WHY 1,600 IS NOT FRIGHTENING

Arithmetic, not a distributed-systems project

Workers claim a shard with a database advisory lock, **no orchestrator, no leader election, no per-replica configuration**, and a crashed worker returns its shard the instant its connection drops.

AND WHAT WOULD MOVE IT

Cross-camera GPU batching

Plausibly 3 to 5x per node, taking 1,600 toward ~400. It is an architecture change, not a flag, so it is **named, not claimed**.

What is built, and what is designed, labelled

INTEROPERABILITY

- ONVIF · RTSP · HLS · WebRTC
- OpenAPI, browsable on the running instance
- **GeoJSON** for everything geographic
- Kafka wire protocol
- One provider interface per watchlist source

No proprietary interface between our own layers.

SECURITY: BUILT

- Credentials **never** in the registry; a URL with credentials embedded is rejected with **422**
- Role- and department-scoped access derived from **registry ownership**, data-driven, not a code path per department
- **Immutable audit** of every view, search, query and export
- A **case reference** binding the purpose of each query

SECURITY: DESIGN

- Managed secret vault with rotation
- mTLS between services
- Camera-VLAN segmentation
- Active-passive DR

Named, not implied. A proposal that blurs built and specified cannot be trusted on either.

Privacy under the DPDP Act 2023. Metadata-by-default *is* the primary control: the platform's default state is that no personal video has moved anywhere. Facial recognition ships **disabled**, requires per-query case binding and audit to enable, and is demonstrated only on synthetic faces.

EXPECTED OPERATIONAL BENEFITS AND IMPACT ON POLICING

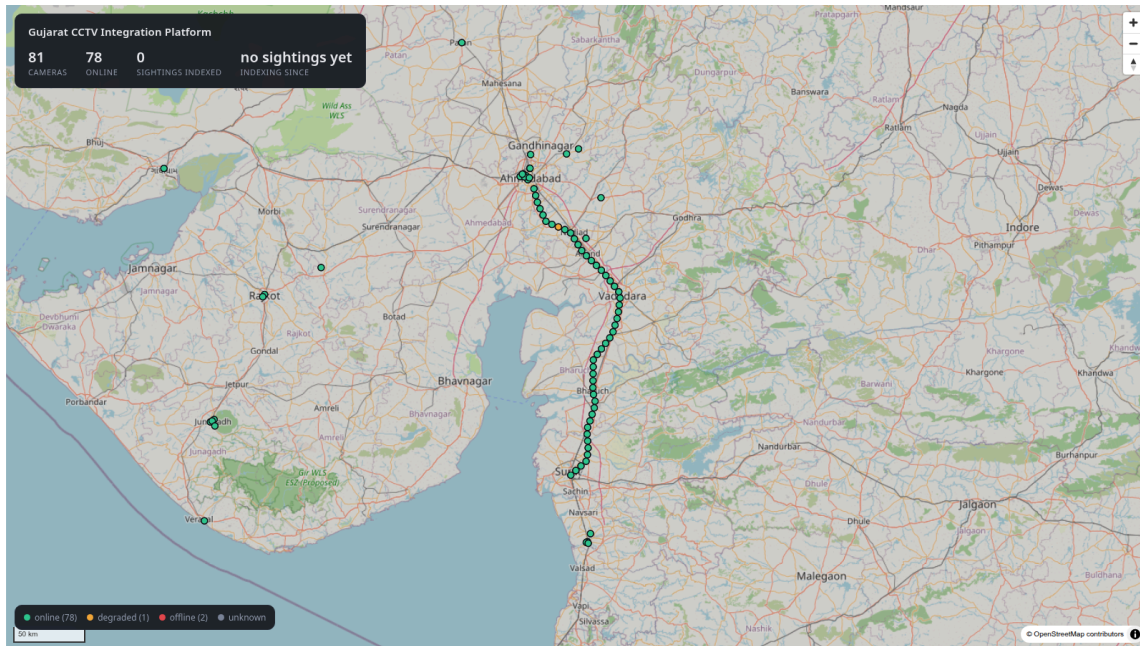
What changes in how the work is done

	TODAY	WITH THIS PLATFORM
Time to locate a suspect vehicle	Camera-by-camera manual review, department by department, over hours	A plate query returns a timestamped route, 26 ms p50, 31 ms p95
Cross-department visibility	A Home Department investigator sees nothing from RTO or Food & Civil Supplies cameras	The registry makes coverage discoverable , you can learn a camera exists before negotiating access to it
Posture	Reactive: footage is reviewed after the incident	Proactive , a hit surfaces while the vehicle is still moving. 1.04 s
Manpower	Operators watch video walls	Operators triage prioritised alerts that carry their own evidence bundle
Infrastructure planning	Where the next camera goes is a judgement call	Gap analysis over the road network . Measured on NH-48: 1.27% of 237 km covered, 51 gaps, largest 4.76 km

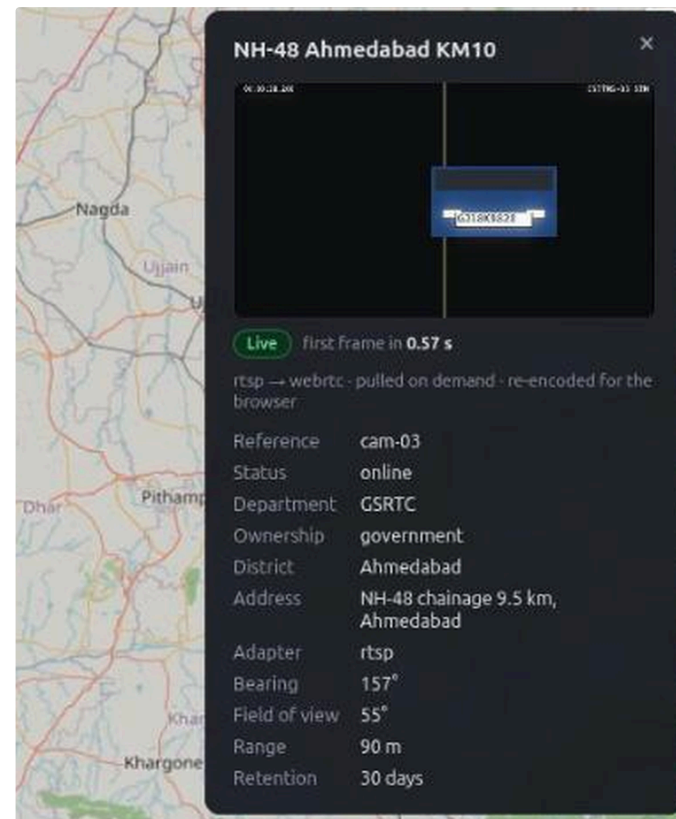
The structural impact: the State gains a searchable index of vehicle movement that already contains the answer to a question nobody has asked yet. **And a cost impact:** ~26 PB and 240 Gbit/s of central infrastructure is **not purchased**, and 26 departments' existing investments keep running.

NOT A MOCK-UP

81 cameras onboarded statewide, every one resolved from the government catalogue



Registry and GIS console. Positions resolve from the catalogue's place names through a gazetteer, never from a camera number, because the reference says camera ids can change. Health is probed, so a login page does not count as online.



Live view, pulled on demand. First frame in 0.57 s. The relay starts when an operator opens the camera and is reaped 45 s after the last viewer, so an unwatched camera costs only its metadata.

MEASURED END-TO-END PERFORMANCE

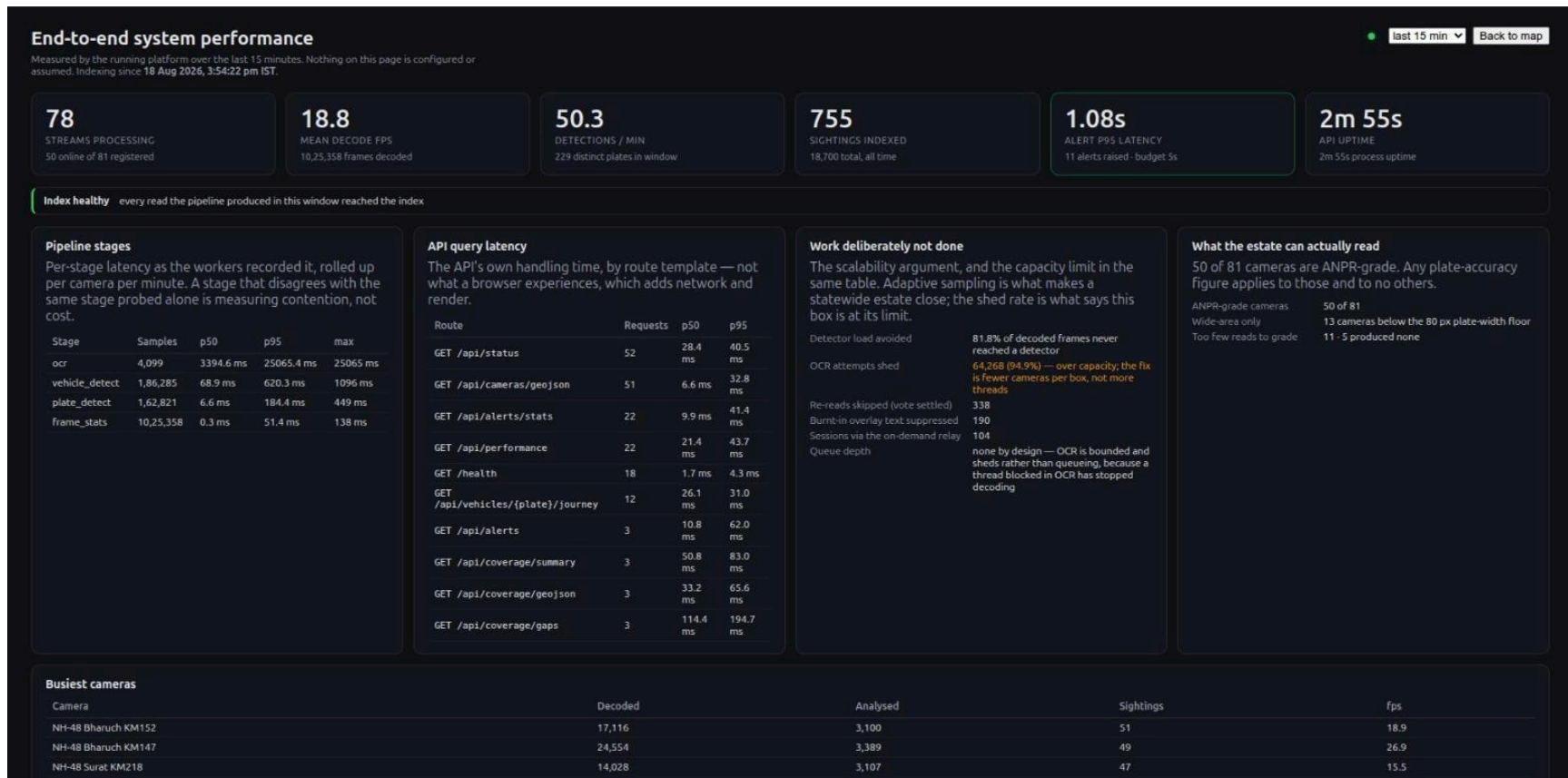
Every number here is live in the platform's own UI, not a rehearsal figure

	BUDGET	MEASURED	HEADROOM
Vehicle trace across the state	2 s	26 ms p50 · 31 ms p95	77×
Detection → alert	5 s	1.04 s	4.8×
Camera onboarding	30 s	5.1 s	5.9×
Click → live video, cold camera	n/a	0.23 s	n/a
Concurrent streams held	50	50 for 15 min, 0 restarts	n/a
Throughput	n/a	104 sightings/min across 71 cameras	n/a
Plate accuracy, ANPR-grade	n/a	82.4% exact · 95.1% within 2	n/a

869 automated tests. Every figure traces to a dated entry recording the method that produced it, which is what makes it checkable rather than asserted.

THE PLATFORM MEASURES ITSELF

The previous slide is a screen in the product, not a number we typed into a deck



The third panel is the one to read. It reports work deliberately not done: 94.9% of OCR attempts shed at 76 cameras on a 20-core box. That is the capacity limit stated as a measurement, and it is why the honest sizing figure is 50 to 57 cameras per box rather than 76.

COMMON EVALUATION AREAS

The seven criteria, and where the evidence for each one sits

	CRITERION	EVIDENCE IN THIS SUBMISSION
1	Successful test case Onboarding and operation on the government feed	81 cameras onboarded by catalogue reconciliation, 30 of them government. Live view and recorded playback both demonstrated; ANPR output written to the index.
2	Solution presentation Problem understanding, model, justification	This deck. Element 1 states the model and why Model 4 was rejected with the arithmetic; slide 2 states the constraint the whole design follows from.
3	Solution architecture Soundness, feasibility, security, interoperability	Six-layer HLD, the workflow diagram on slide 9, and the interoperability and security slide. Full design in the submitted HLD document.
4	Working platform and demonstration	Slides 19 and 21 are screens from the running platform. Demonstrated on both our own simulated estate and the government feed.
5	Video analytics output ANPR, detection, timestamps, reports	ANPR with per-condition accuracy, vehicle detection and tracking, tamper detection, appearance attributes, and CSV and PDF detection reports carrying the evidence crop.
6	Scalability and PoC readiness Toward 80,000 cameras	Measured capacity per box, sharding by advisory lock with no orchestrator, and the shed rate published rather than hidden. Slides 16 and 21.
7	Submission completeness	Deck, HLD, workflow diagram, demonstration video, public repository, live instance with a viewer account, and API documentation.

Nothing in this column is aspirational. Where a capability is designed rather than built, it is labelled as such on slide 17, and the limits we know about are stated on the final slide rather than left for an evaluator to find.

BONUS CONSIDERATIONS

Each of the six, and what we built against it

NOVEL HYBRID ARCHITECTURE

All three models, not a choice between them

Registry as control plane, direct feed access for analytics, adapters for everything that does not fit. The registry is mandatory; the other two are how it earns its keep.

CROSS-CAMERA CORRELATION

Route reconstruction in 26 ms

Road-snapped through OSRM, with per-hop plausibility. An impossible hop splits the journey and flags a probable cloned plate rather than being discarded.

ANALYTICS BEYOND ANPR

Four more, all shipped

Vehicle detection and tracking, camera tamper detection, appearance attributes for feeds that cannot resolve a plate, and face detection as a disabled plugin proving the pipeline takes modules.

EDGE AND BANDWIDTH

About 0.5 kbit/s per camera

Analytics run beside the camera and only metadata crosses the link. Video is pulled on demand and released 45 s after the last viewer, so cost is per viewer, not per camera.

SECURITY AND PRIVACY

Audited, role-split, crops not frames

Every view, search, trace and export is recorded in an append-only log. Two roles, and the evaluation account is read-only by API enforcement. One 8 kB evidence crop per sighting; no recorded video.

DASHBOARDS AND APIS

Integration-ready by default

GIS console, alert desk, journey explorer and a live performance page. Every capability is an OpenAPI-documented endpoint, with GeoJSON for anything geographic.

A proposal that lists only strengths cannot be checked

- **All 30 government cameras fall below ANPR grade, and none of the 50 simulated ones do**, the platform grades this itself, from crops it measured. Their crops average **66 px** across, about 7 px per character. They are correctly deployed situational-awareness views. Every accuracy figure here is **scoped to ANPR-grade cameras**. Measured live on 14 Sep with the learned detectors: 344 real vehicles in 10 minutes, boxes on real plates, **median plate width 29 px** and **0 valid plates** from any reader we tested; the report lists detected vehicles with type, colour and time.
- **Night accuracy was 58.6% exact with Tesseract**. docTR, now the default, finds 62% of night plates against 35% on the same clips.
- **A vehicle id joined by appearance is a lead, not an identification**. 43 links checked by eye on the government cameras: 34 right, 7 wrong, 2 unclear; at the shipped threshold 16 of 17 were right. **None crossed cameras**, so there it re-finds a vehicle on one camera rather than following it between cameras. The plate trace never follows an appearance link.
- **Edge autonomy, push notification routing, mTLS and the vendor VMS adapters are designed, not built**. They are in the technical proposal because the architecture is incomplete without them, and labelled because a proposal that blurs built and specified cannot be trusted on either.
- **Watchlist data is representative, not live**. No team has live VAHAN access for a hackathon.
- **The 1,600-node figure assumes traffic like the measured corridor's**. A quiet road costs less, a toll plaza more, and the platform reports sightings-per-camera-per-minute *per camera*, so the model stays checkable at rollout rather than only at design time.